

CSS 是什麼？

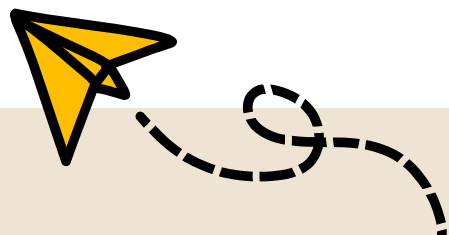
Cascading 📎 階層式的；套用多種(層)

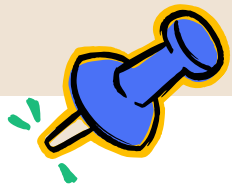
Style 📎 樣式；外觀呈現的樣子

Sheet 📎 表(自成一格)；有固定的格式

HTML 📎 結構，顯示內容；看到

CSS 📎 外觀，呈現視覺的變化；好看



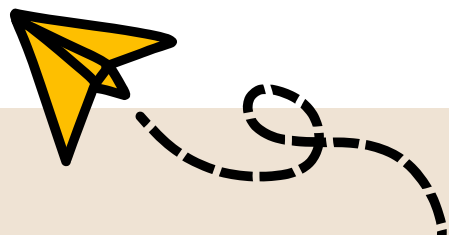


css **BOX** model

HTML element  `<div>` ; `<img>` ; `<a>`...

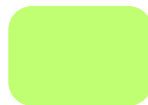
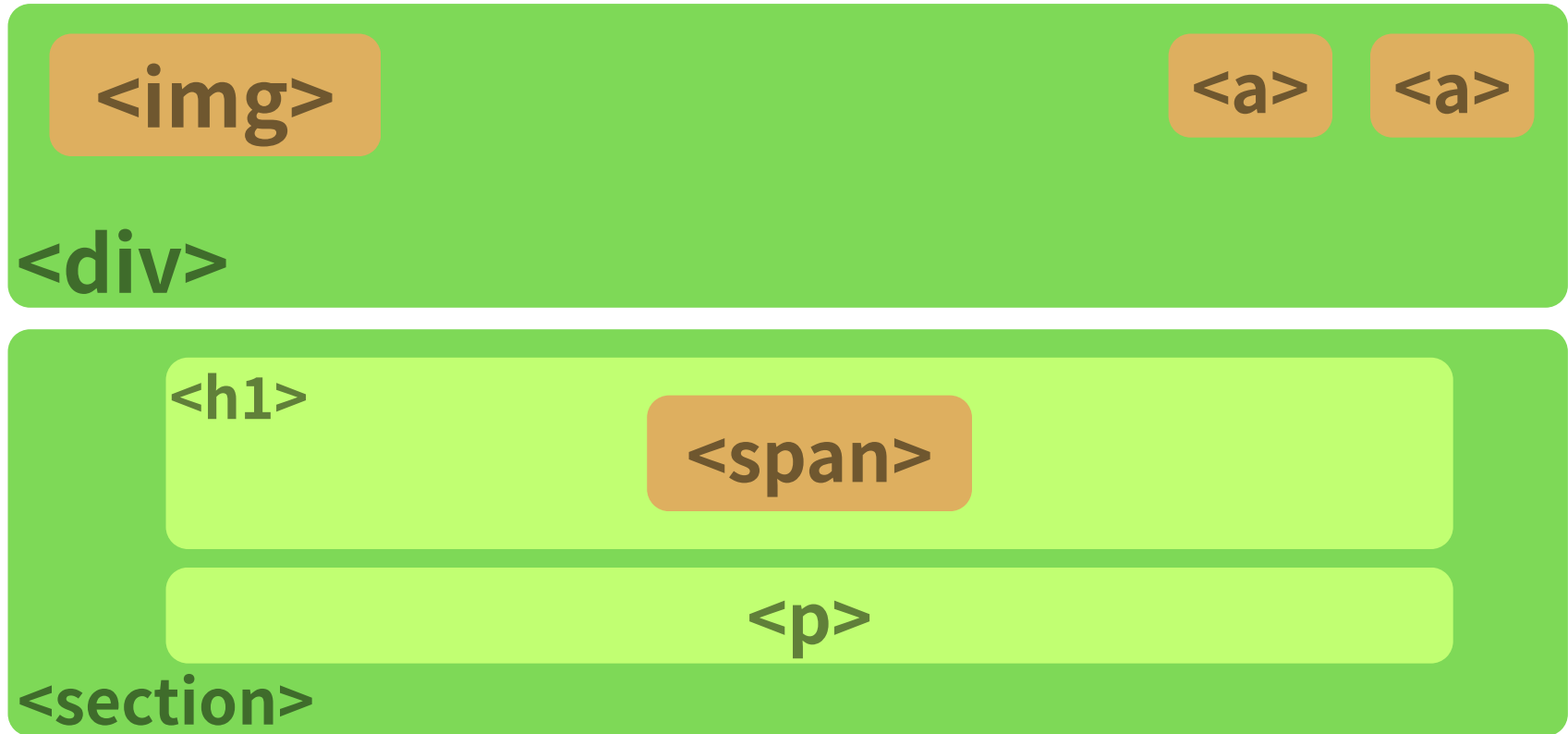
CSS把HTML element放入(看不見)的**BOX**

BOX  Block 或 Inline





css **BOX** model



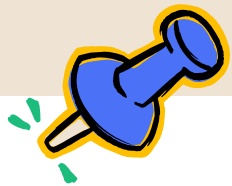
Block



Inline

Goback





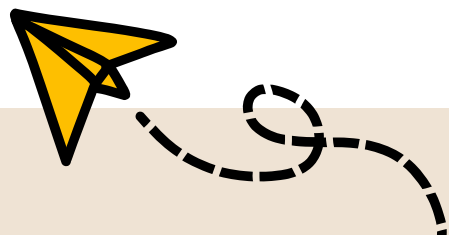
css **BOX** model

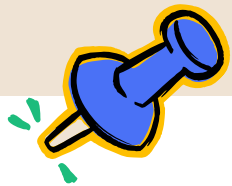
CSS處理BOX的「部位」

BOX的內容  font-size, color...
字型，色彩...

BOX的本體  width, border, margin...
寬高，框，邊距...

BOX的位置  Flexbox, Grid, Position
網格，位置...



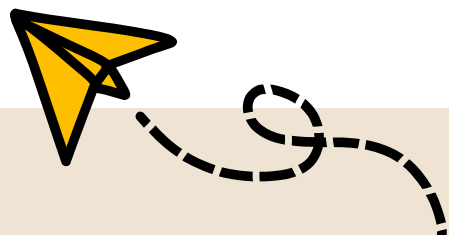


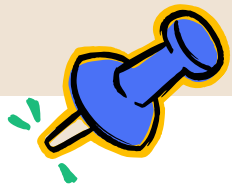
CSS 「懸念」

CSS的內容 → 比HTML，多很多，持續成長...

CSS的理論 → 必須搞懂原理...

CSS的繁鎖 → 應付不同設備的Layout





CSS Rule(規則：寫法or格式)

Property
屬性

Value
值

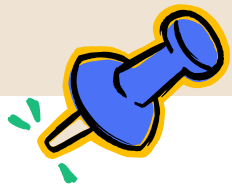
font-size : **60px** ;

color : **blue** ;

冒號(隔開)

分號(結束)



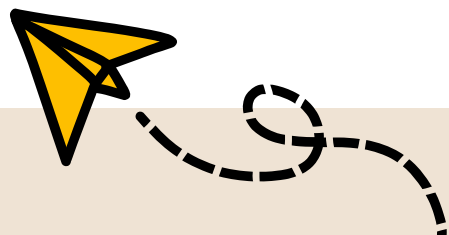


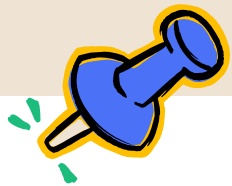
CSS Rule(規則：寫法or格式)

font-size : 60px;

color : blue;

Declaration
宣告，聲明





CSS Rule(規則：寫法or格式)

宣告的對象



大括號

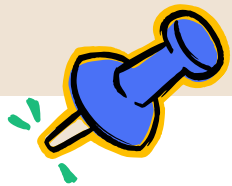
{

font-size : 60px;
color : blue;

}

Declaration block
宣告區塊





CSS Rule(規則：寫法or格式)

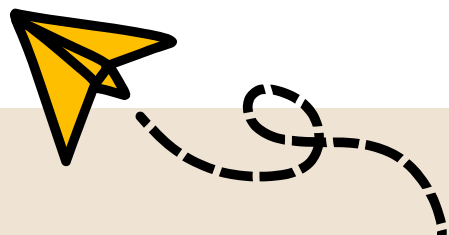
Selector
選擇器

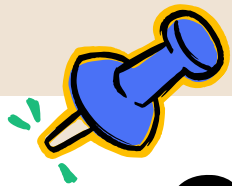
關卡

h1 {

**font-size : 60px;
color : blue;**

}





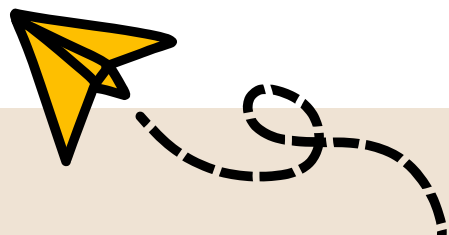
CSS 進場的方式

寫在 HTML(不建議)

1.inline **NO!**

2.Internal

3.EXTERNAL





CSS Text styling >>>>

font-weight: 100 200 ~ 900;

標題：500~900；其它：300~400

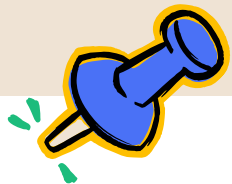
font-style: italic;

normal, italic, oblique 10deg

○預設

○設定斜體的角度
與印刷較有關





CSS Text styling

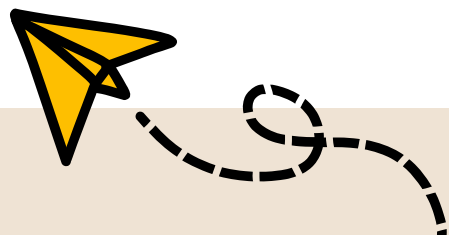
常用設定，移除的前置

`list-style: none;`

`none, disc, circle, square, decimal`

ul 的預設

ol 的預設





CSS Text styling

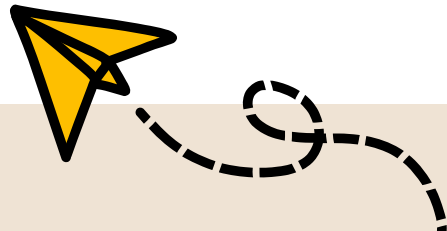
text-decoration: ① ② ③;

常用，目的：移除[a](#)的底線

① **none**, underline, overline, line-through

② **named**, RGB, Hex

③ **solid**, dotted, dashed, double, wavy



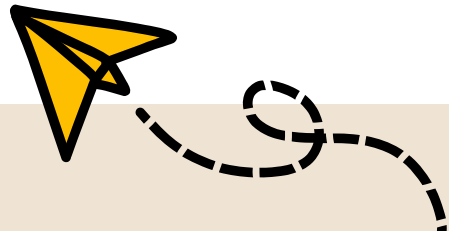


CSS Text spacing>>>>

`text-align: center;`

預設

left, right, center, justify





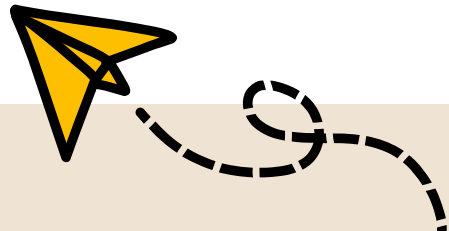
CSS Text spacing

line-height: 1.5;

常用，以「倍數」調整

unitless, pixels, percentages, ems

標題(h1,h2) : < 1.5
一般文字 : 1.5~2





CSS Text spacing

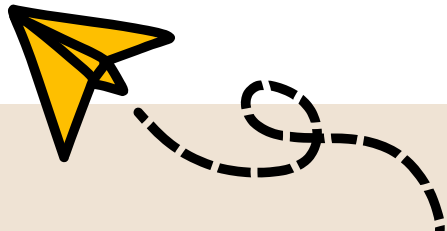
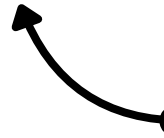
letter-spacing: 8px;

absolute(絕對)



pixels, percentages, rems

relative(相對)





CSS Text spacing

font-size: 16px;

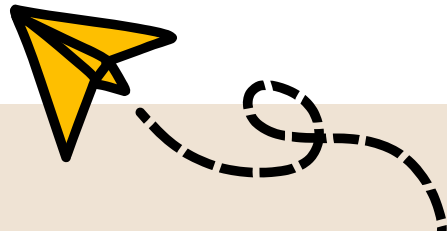
↖ absolute(絕對)

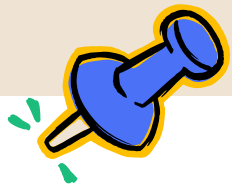
pixels, rems

↖ relative(相對)



Type Scale





CSS font family

typeface(字的面貌)

Serif
(襯線)

Times New Roman

MonoSpace
(等距字體)

Courier New

Display
(展示字體)

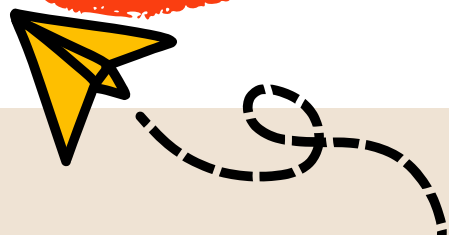
Impact

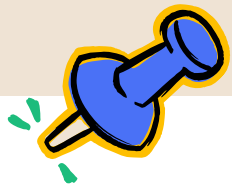
Sans-Serif
(無襯線)

Arial

Cursive
(草書)

Comic Sans MS



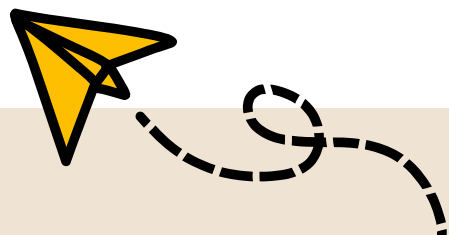


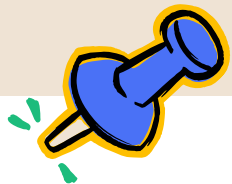
CSS font family

首選，單引號

font-family: ` Segoe UI `, sans-serif;

最終備案



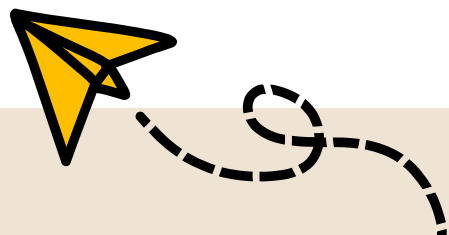


Google Font

- ① 尋找合適的 font
- ② Embed(嵌入)到 HTML
- ③ 在 CSS 增加 font-family



Google Font





Named (色彩名稱)

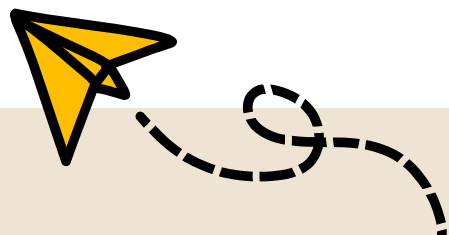
據說：有140種以上



Red

Green

Blue





RGB (紅綠藍)

rgb(Red, Green, Blue)

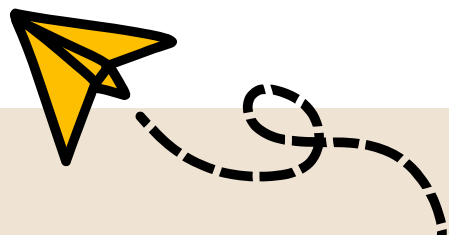
0~255，共有256階

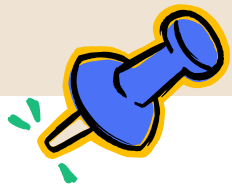
👉 rgb(152, 116, 249)

rgba(Red, Green, Blue, Alpha)

👉 rgba(120, 85, 60, 0.5)

0~1，建議 0.1~0.9





#HEXcode (16進制色碼)

#RedGreenBlue

0~9;A(10)~F(15)

👉 #121212

👉 #AB6235
❌ 無法縮寫

#HEXcode 「縮寫」

👉 #11CCDD
👉 #1CD ✓

👉 #AA6633
👉 #A63 ✓

👉 #AA1234
❌ 無法縮寫

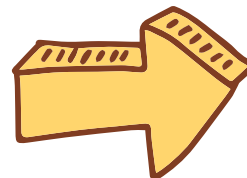




Shades of GRAY (灰階色調)

rgb(0, 0, 0)

#000000



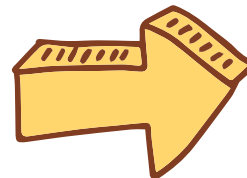
#000

rgb(64, 64, 64)

#404040

rgb(136, 136, 136)

#888888



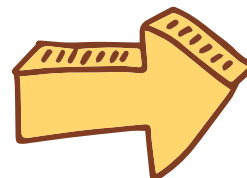
#888

rgb(192, 192, 192)

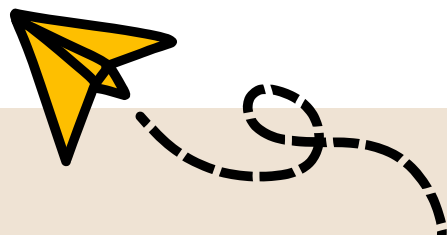
#C0C0C0

rgb(255, 255, 255)

#FFFFFF



#FFF





Primary Secondary Color



[Open Color](#)

好站
報報



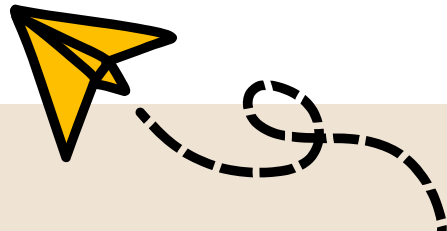
[Tint and Shade](#)

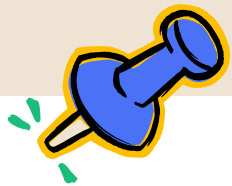


[Color Picker](#)



[HTML Color Codes](#)





Brand Guideline(品牌指南)

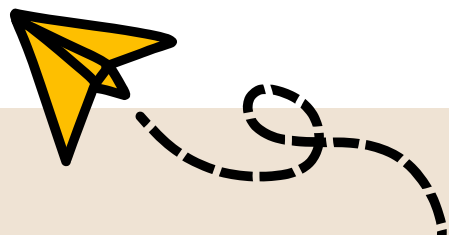


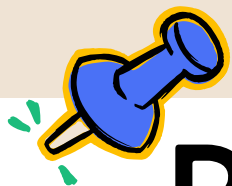
Duolingo



Slack

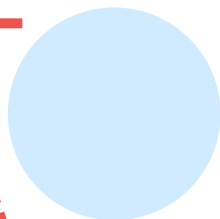
好站
報報





Primary Secondary Grey

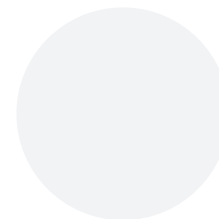
Tint



#d0ebff

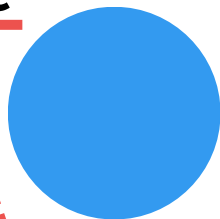


#e5dbff

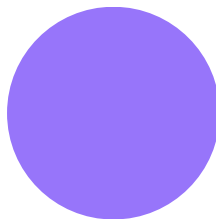


#f1f3f5

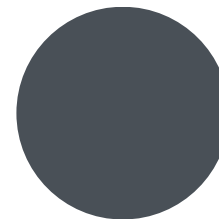
Base



#339af0

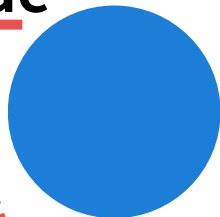


#9775fa



#495057

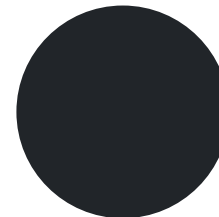
Shade



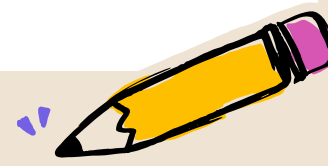
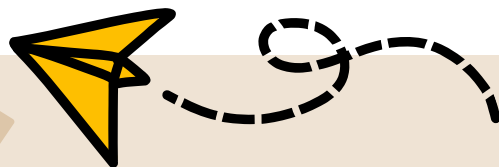
#1c7ed6



#7048e8



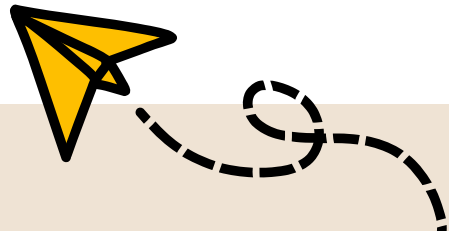
#212529





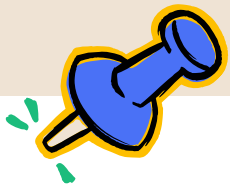
Selector(選擇器) 種類

- ① **Basic(基本)** ➡ `[Elements 或 # 或 .(小數點)]`
- ② **Combinator(組合)** ➡ `[ul > li]`
- ③ **Pseudo-class(偽類別)** ➡ `[a:hover]`
- ④ **Pseudo-element(偽元件)** ➡ `[::before]`



mdn CSS Selector





Basic: Element (Type)

Selector

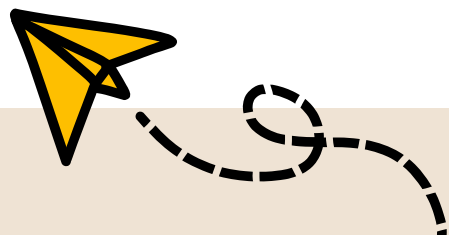
選擇器

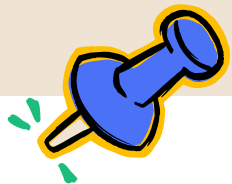


h1 {

**font-size : 60px;
color : blue;**

}





Basic: ID and Class

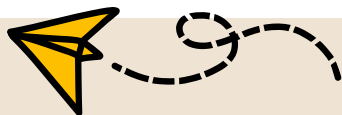
`<p id=" idd" >id 段落文字</p>`

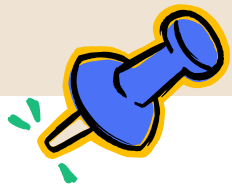
`<p class=" cla" >class 文字</p>`

`<p>class 文字</p>`

`#idd {
 color : blue;
}`

`.cla {
 color : green;
}`



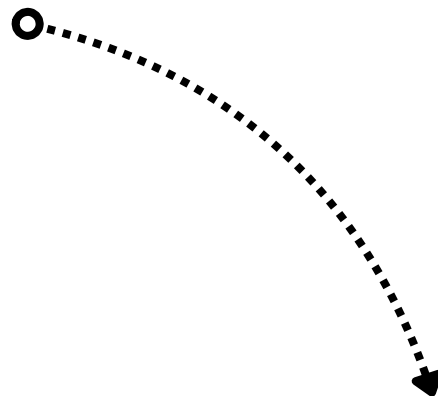


Cascade=Priority (階層=優先)

```
h1 {
```

```
  color: blue;
```

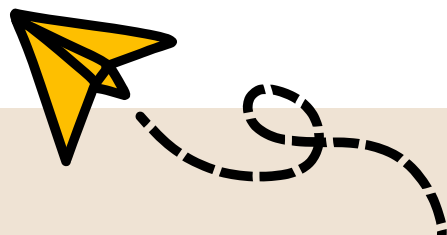
```
}
```

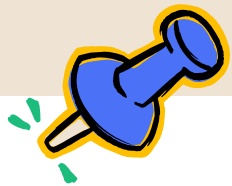


```
h1 {
```

```
  color: red;
```

```
}
```





Cascade=Priority (階層=優先)

<head>

<link~ web.css >

<link~ web2.css >

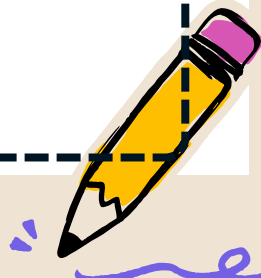
</head>

web.css

```
h1 {  
    color : blue;  
}
```

web2.css

```
h1 {  
    color : red;  
}
```

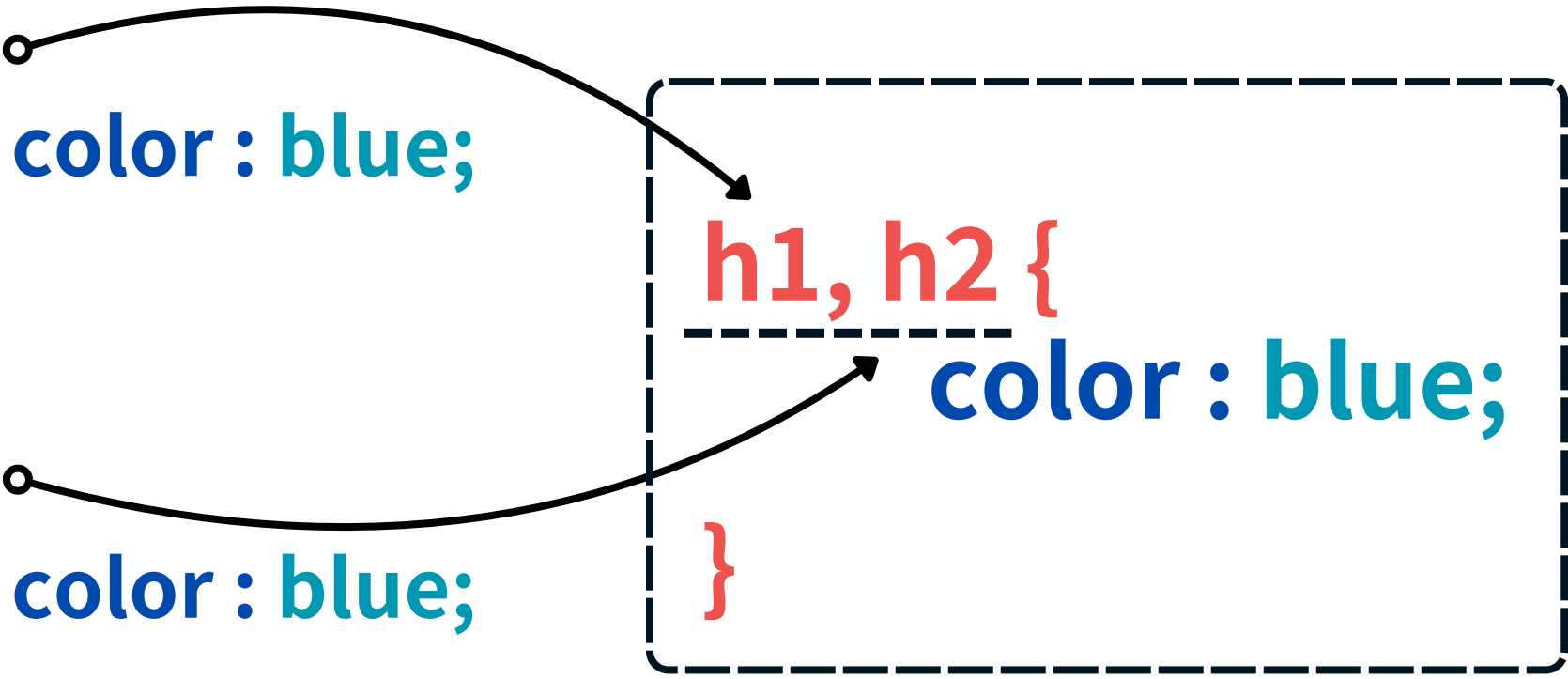




Basic: Grouping

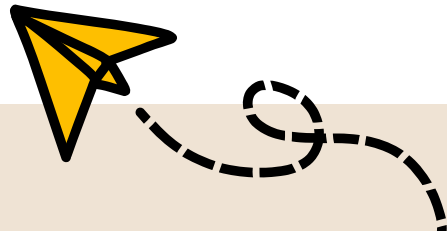
```
h1 {  
  color : blue;  
}
```

```
h2 {  
  color : blue;  
}
```



A dashed rectangular box on the right side of the slide groups the two CSS rules. Two curved arrows originate from the opening curly braces of the 'h1 {' and 'h2 {' rules on the left and point to the 'h1, h2 {' rule inside the dashed box. The text inside the box is 'h1, h2 {' followed by 'color : blue;' and a closing brace '}'.

```
h1, h2 {  
  color : blue;  
}
```





Basic: Grouping(多重class)

```
.btn{  
  font-size: 16px;  
  font-weight: 400;  
}
```

基本設定

可能會重覆，但仍需設定

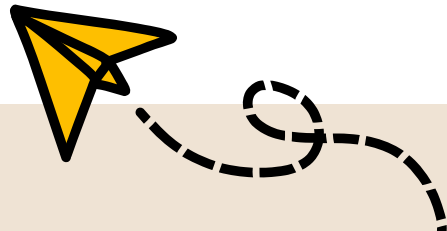
```
.btn-primary{  
  color: blue;  
}
```

主要

```
.btn-secondary{  
  color: green;  
}
```

次要

Component UI





Basic: Grouping(多重class)



空一格

Grouping 多重 class

.btn{
font-size: 16px;
font-weight: 400;
}

基本設定

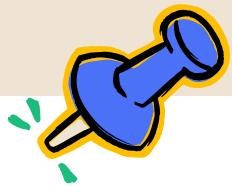
.btn-primary{
color: blue;
}

主要



<http://Bootstrap>





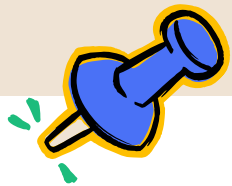
Pseudo class

a: hover

選擇器

Pseudo class





Pseudo: State

User Interaction (使用者互動)

a:link



未點過

a:visited



已點過

} 針對 **<a>**

a:hover



滑鼠懸停在上

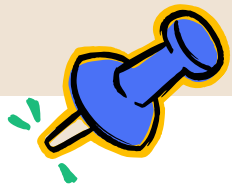
a:active



滑鼠按下未放

} 非針對 **<a>**



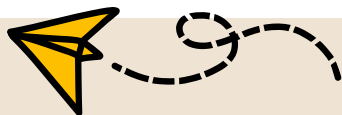


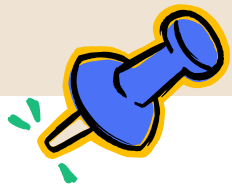
Pseudo: <a> State

Pseudo state


```
a:link, a:visited {  
    color: green;  
}
```

```
a:hover, a:active {  
    color: blue;  
}
```





btn Class + State

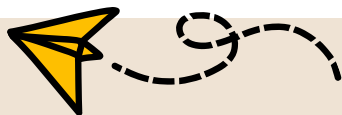
```
<a href="#" class="btn btn-primary" >
```

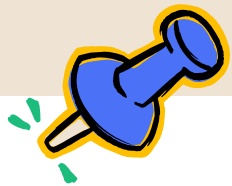
多重 class

```
</a>
```

```
.btn-primary:link,  
.btn-primary:visited {  
  color: green;  
}
```

```
.btn-primary:hover,  
.btn-primary:active {  
  color: blue;  
}
```





btn Class + State

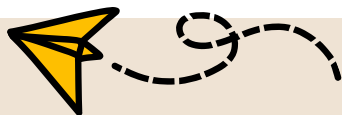
```
<a href="#" class="btn btn-secondary" >
```

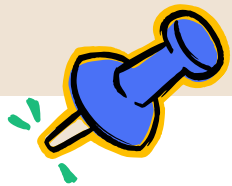
多重 class

```
</a>
```

```
.btn-secondary:link,  
.btn-secondary:visited {  
  color: green;  
}
```

```
.btn-secondary:hover,  
.btn-secondary:active {  
  color: blue;  
}
```





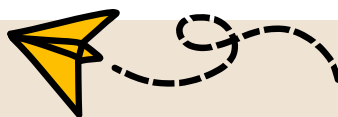
Combinator(組合) selector

div p  Child(後代)

div > p  Direct Child(直接後代)

div + p  Adjacent sibling
(相鄰層級)

div ~ p  Sibling (兄弟或姐妹)



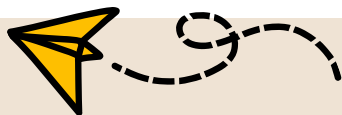


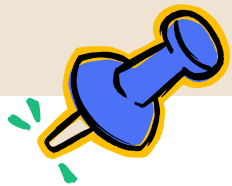
Conflicting vs Specificity

(衝突) (權重)

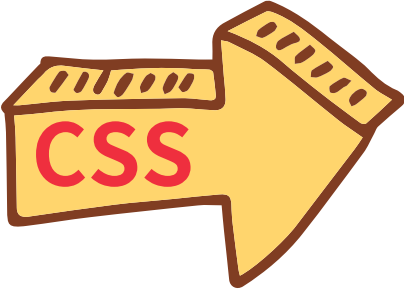
以 **Type(element)** 為「全域」作設置

再以 **Class** 覆蓋，為個別作設置



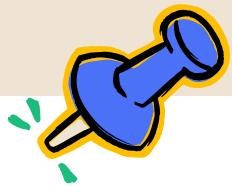


Inheritance(繼承)

Parent(父母)  Children

Parent(父母)  Children 





Inheritance 與 Property

color

text-align

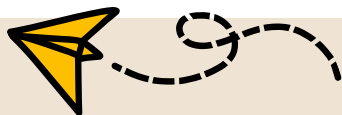
line-height

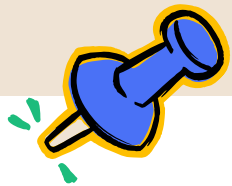
font-size

font-family

font-weight

letter-spacing





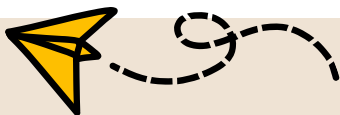
Inheritance 與 Container

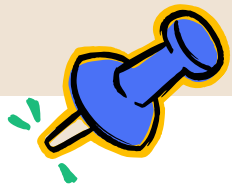
<div>

<h1>

<p>

<a>



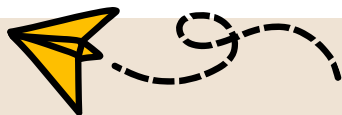


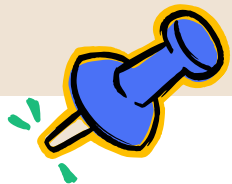
Inheritance 與 body

<body>

所有 <TAG>

設置：相同，共用



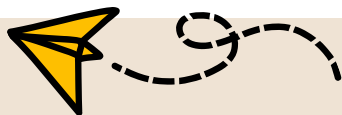


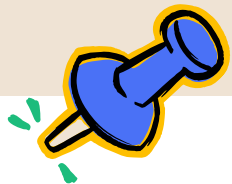
Universal(通用) selector

* { 設定宣告 }



包含<body>的所有設定





body 與 Universal(通用)

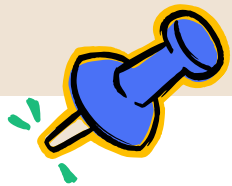
<body>之內

所有 <TAG> 的共用設定

整個 <html>

包含<body>的所有設定



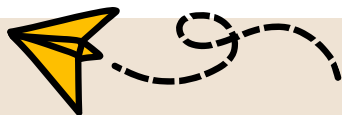


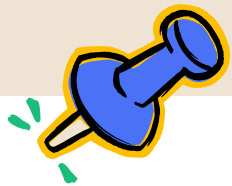
Universal(通用) selector

* { Resetting 宣告 }



非 套用所有設定



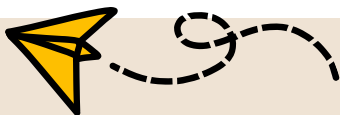


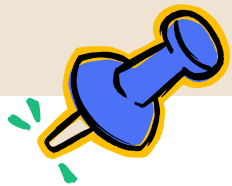
body selector

{**宣告** 文字共用屬性}



非 套用所有設定



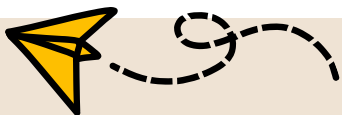


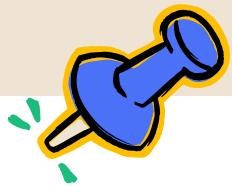
Pseudo element(偽元件)

h2::first-letter

Pseudo element

選擇器



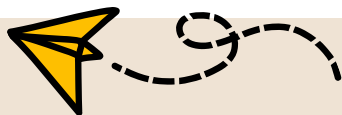


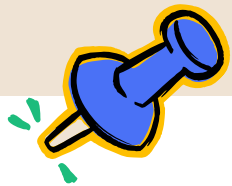
Pseudo element(偽元件)

文字樣式

::first-letter

::first-line

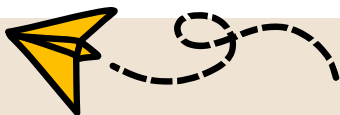


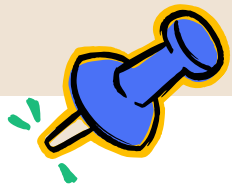


Pseudo element(偽元件)

選取

::selection





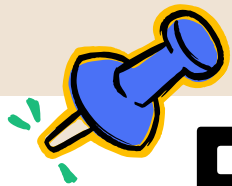
Pseudo element(偽元件)

內容置入

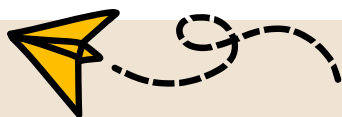
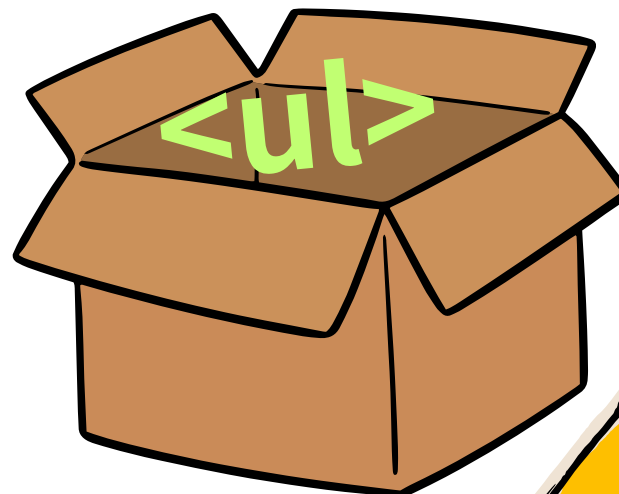
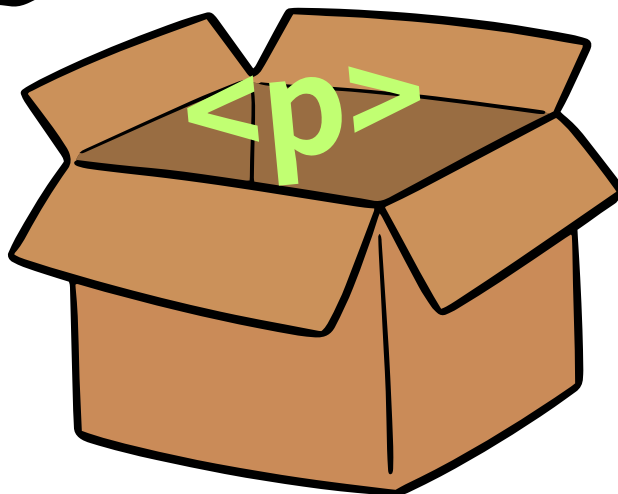
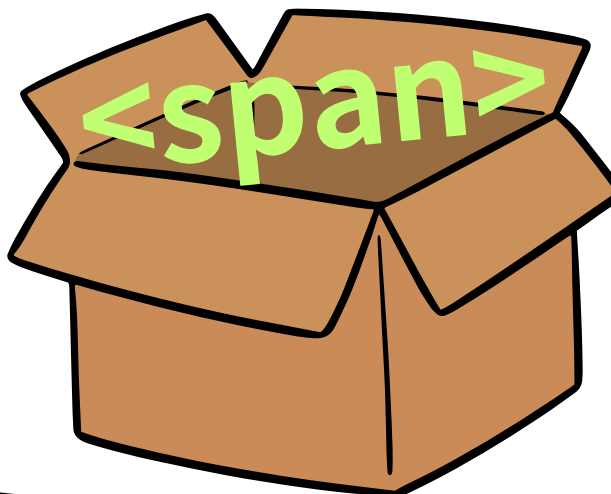
::before

::after



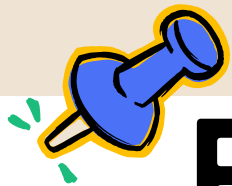


Box Model

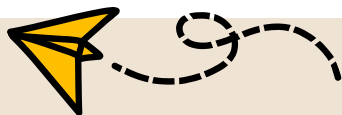
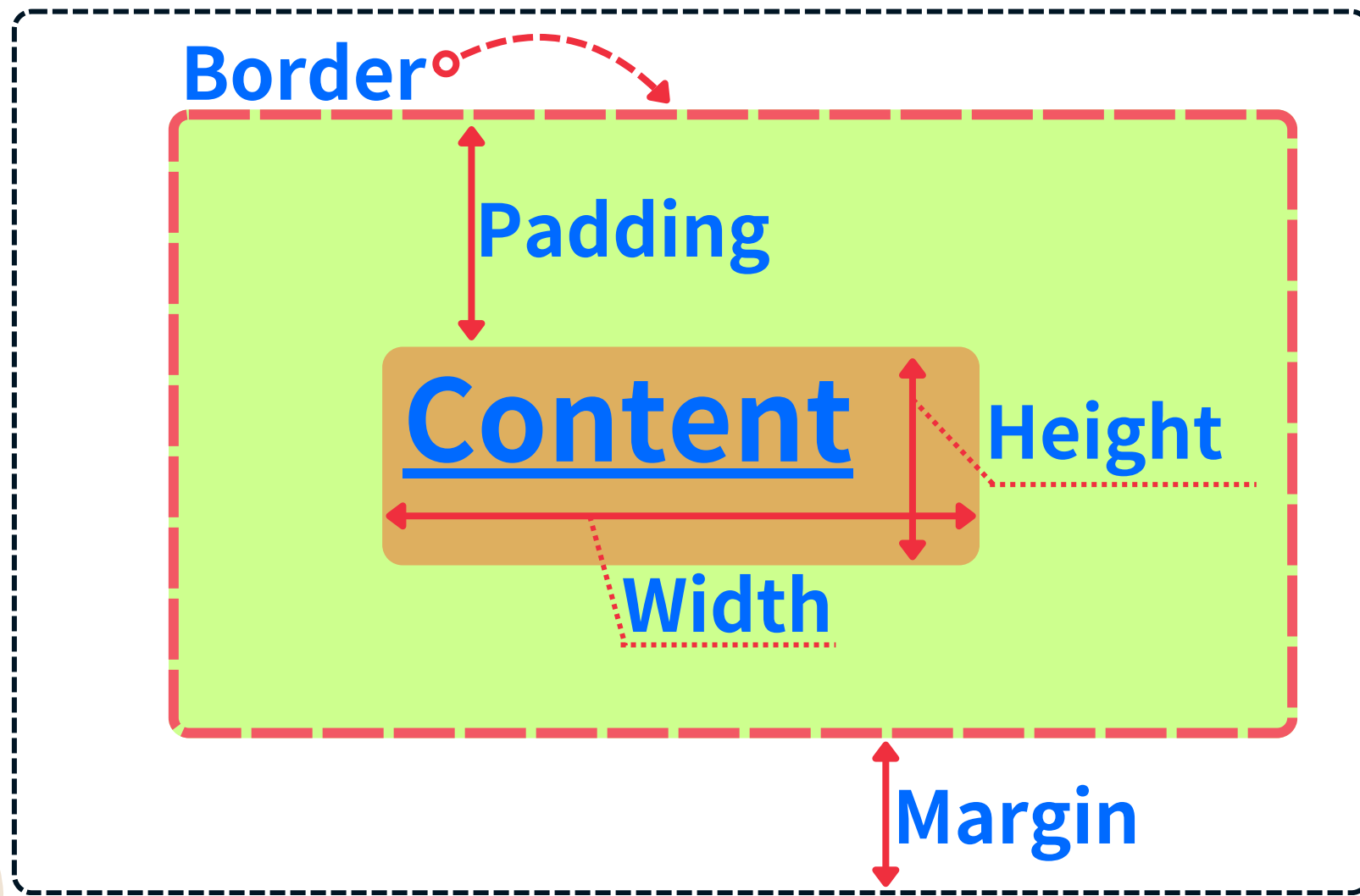


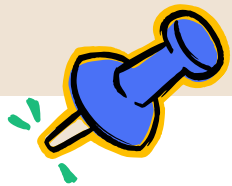
Goto





Box Model

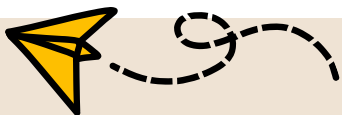


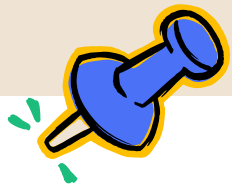


Block

`<h1>`
`<p>` }

font-size決定Height





Block

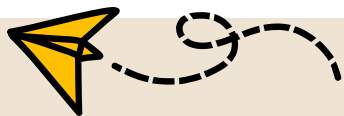
font-size決定**Height**

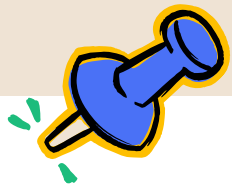


User Agent 的影響



line-height



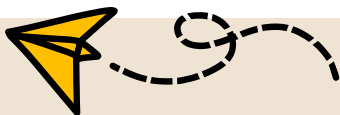


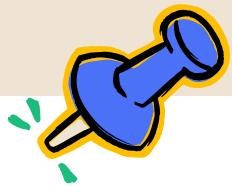
inline



`<a>` font-size決定W與H

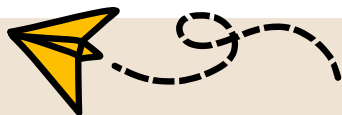
`<img>` 自訂Width,Height

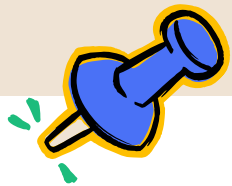




Container

font-size
padding } 內容, 決定 Height





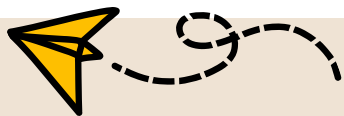
Container與Element之Height

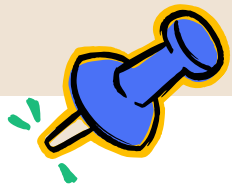
font-size
padding
margin



決定Height

適應不同螢幕(解析度)



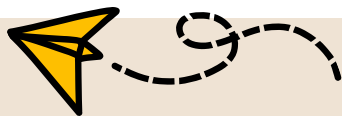


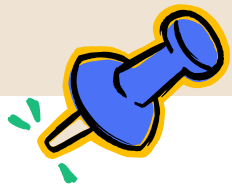
Container與Element之Width

Container 決定 Element

Layout 決定 Container

適應不同螢幕(解析度)





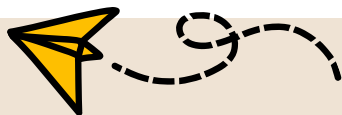
inline Element 的 W與H

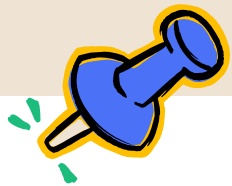
可以使用

img
input
select
button
textarea

不可以

{ a
span





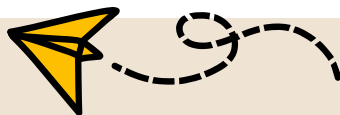
Longhand 與 Shorthand

長名稱
(Longhand)

padding-top
padding-right
padding-bottom
padding-left

短名稱
(Shorthand)

padding





Longhand Idea



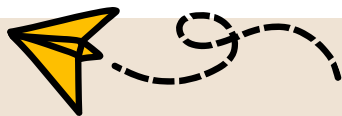
padding-top = 20px ;

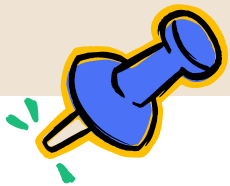
padding-right = 20px ;

padding-bottom = 20px ;

padding-left = 20px ;


**BRIGHT
IDEAS**





Shorthand



padding = 全部 ;

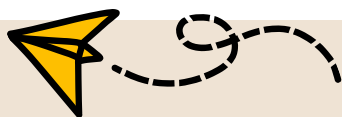
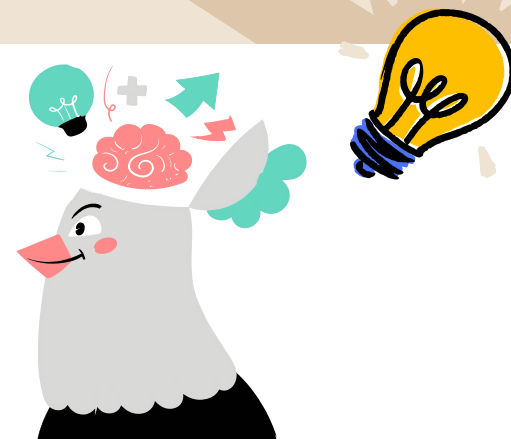
padding = 上下 左右 ;

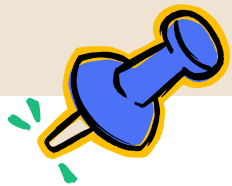
padding = 上 左右 下 ;



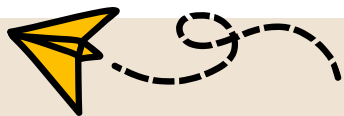
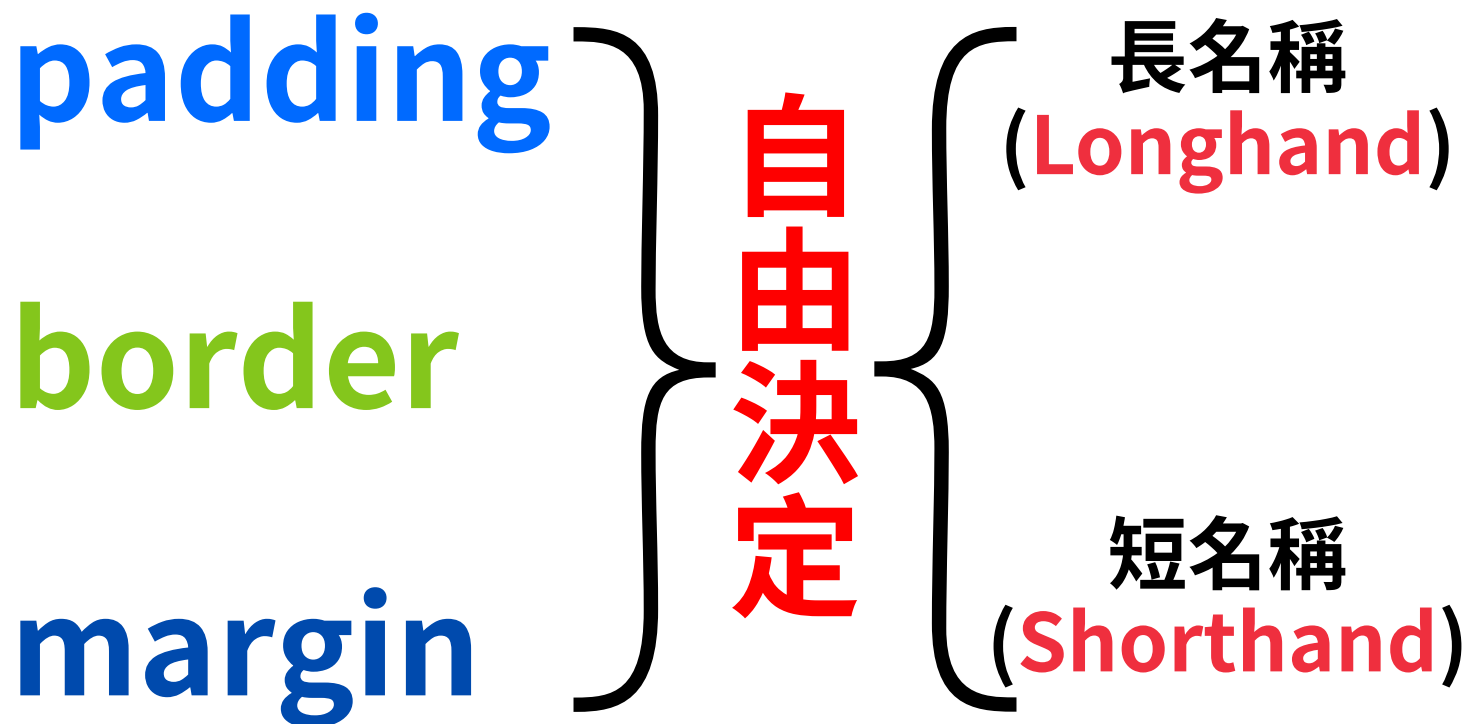
腦袋轉

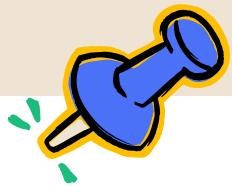
padding = 上 右 下 左 ;





Longhand 與 Shorthand



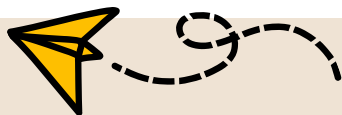


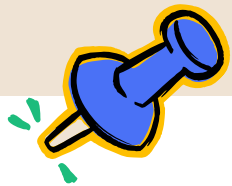
Border (Longhand-全部)

border-width

border-style { solid double
dashed dotted

border-color





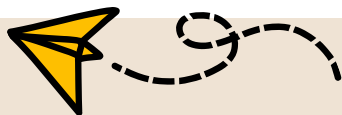
Border (Longhand-單邊)

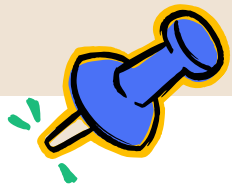
border-top

border-right

border-bottom

border-left

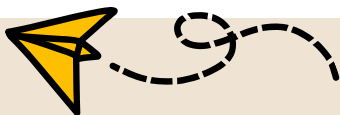


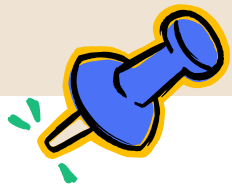


Border (Shorthand-全部)

border : **style** **width** **color** ;

不能省略





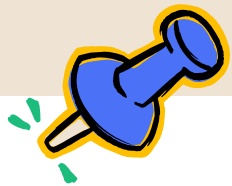
Border 預設(無設定)

style ➡ none

width ➡ 0px

color ➡ element color



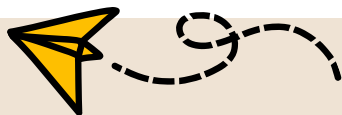


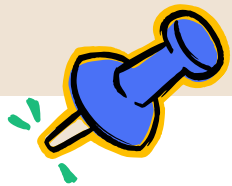
Border Radius

border-radius = 10px ;

Width 或 Height

二分之一 是 MAX





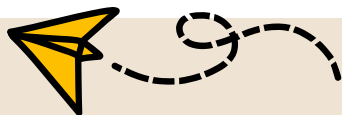
Box sizing

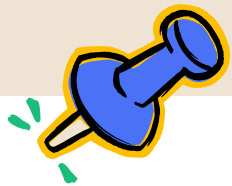
content-box

外擴型

border-box

內縮型





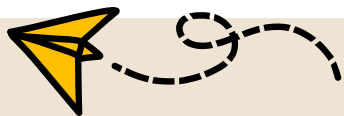
Box sizing (content)

內容的總和就是 Box 的尺寸

內容
總
和

Width
+
Height
+
Padding
+
Border

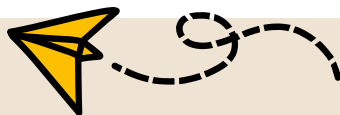
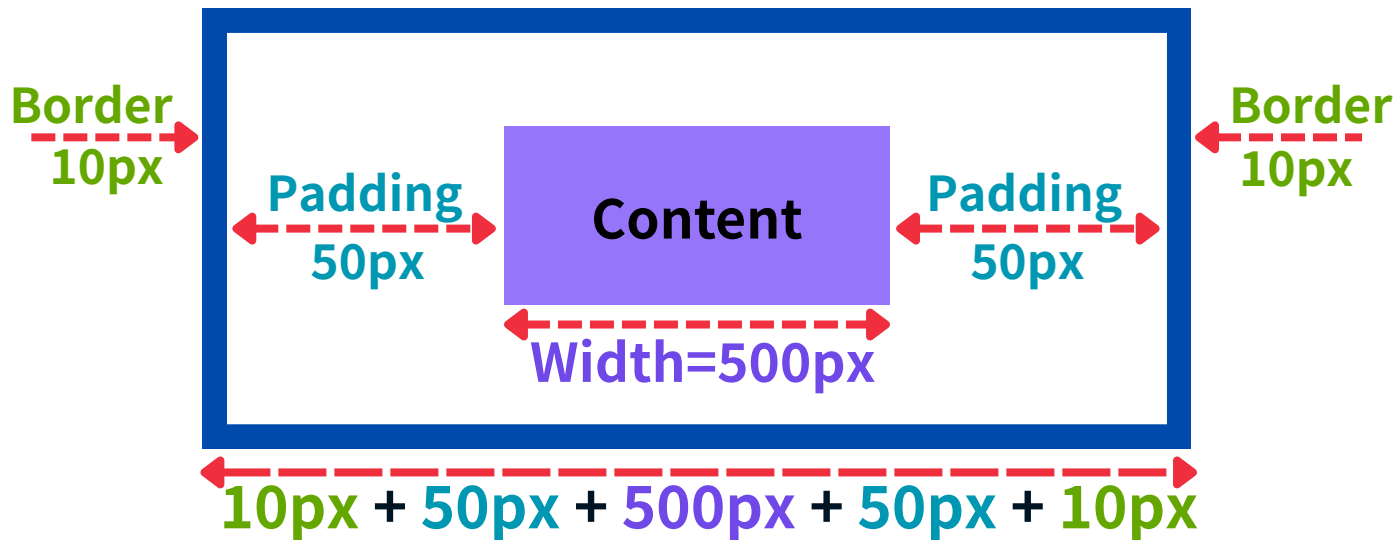
BOX
尺寸





Box sizing (content)

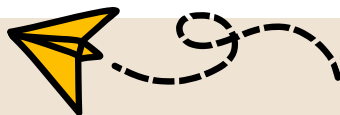
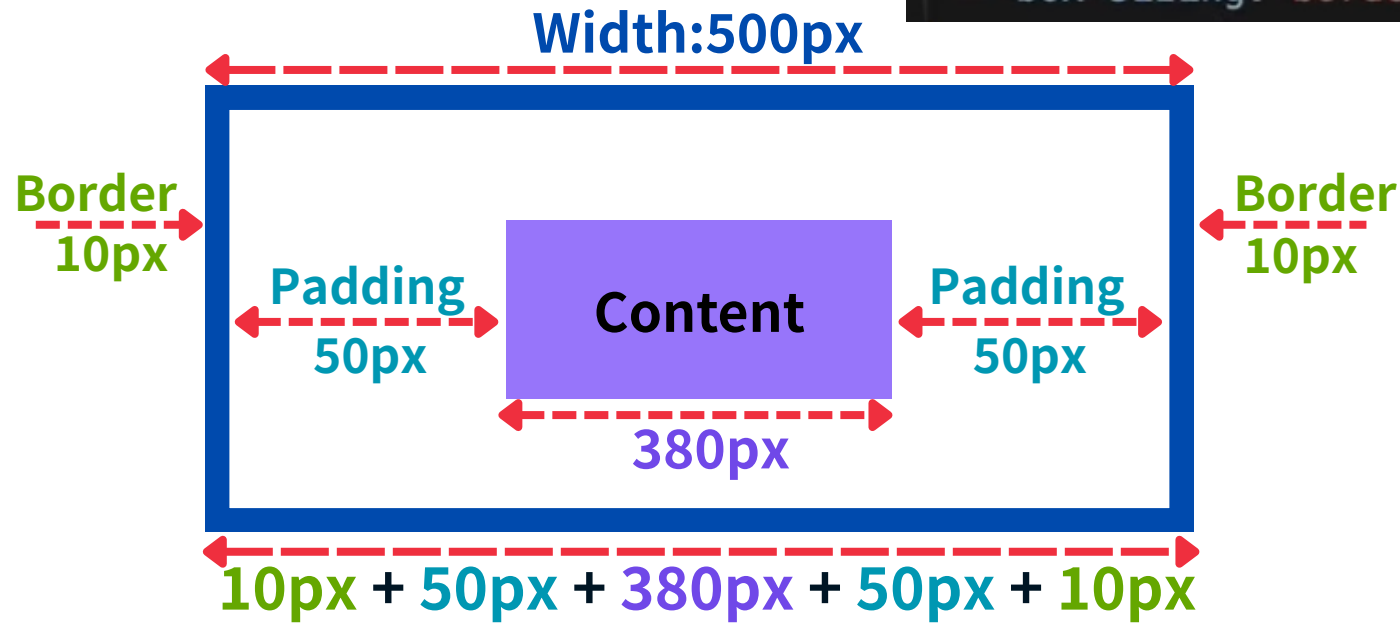
```
.btn {  
  width: 500px;  
  height: 200px;  
  padding: 50px;  
  border: 10px solid blue;  
}
```

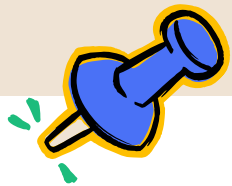




Box sizing (border)

```
.btn {  
  width: 500px;  
  height: 200px;  
  padding: 50px;  
  border: 10px solid blue;  
  box-sizing: border-box;  
}
```





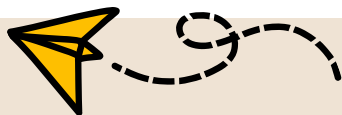
Box sizing

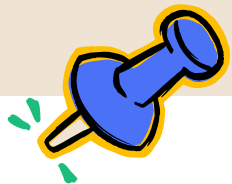
box-sizing: content-box ;

預設，內容優先

border-box

常用，BOX優先



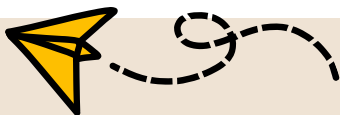


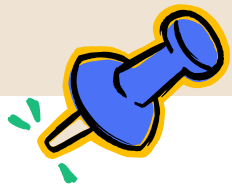
BLOCK 受控於 Box model

BLOCK
element

可以控制

width
height
padding
border
margin



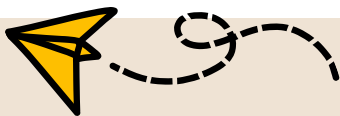


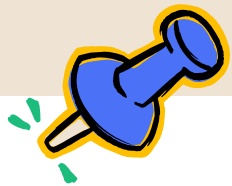
inline不完全受控於 Box model

Inline
element

控制不完全

width
height
padding
border
margin





Browser 預設 CSS

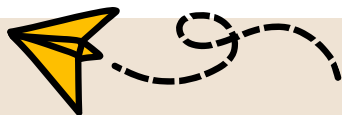
`<body>`

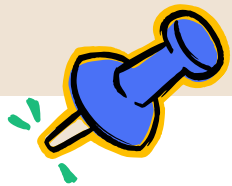
margin 8px

`<ol>` `<ul>`

margin
top & bottom
16px

padding left
40px





Browser 預設 CSS

<h1>

margin
top & bottom
21px

font-size
32px

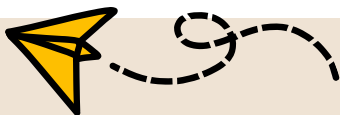
font-weight
700

<p>

margin
top & bottom
16px

font-size
16px

font-weight
400



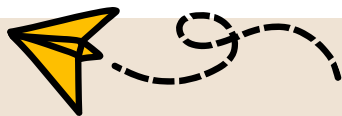


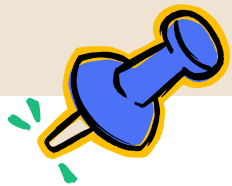
Reset CSS

***** {
margin 0px
padding 0px
box-sizing
border-box

a {
text-decoration
none

ul, ol {
list-style
none





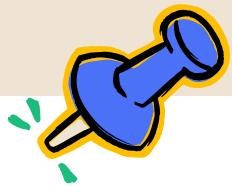
Main Container

將<body>裡的內容放入
main container並置中

margin : 0 auto ;

 設寬不設高，就是 auto





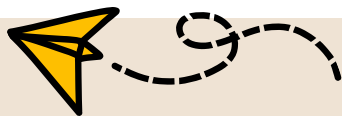
固定 與 最大

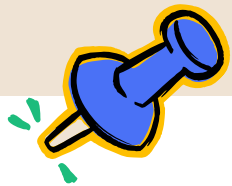
Width 固定寬

固定；不會被改變

Max-width 最大寬

最大；可小，不會再大



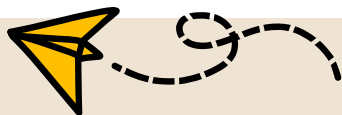


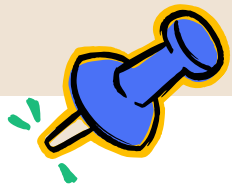
block與inline (Regular與Replaced)

Block : W&H 可控 , 例 : <div>, <h1>

Regular inline : W&H 不可控 ---> display: inline-block
W&H ; 才可控

Replaced inline : W&H 可控 , 例 :

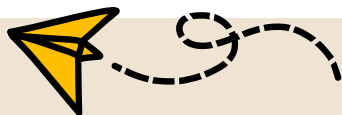


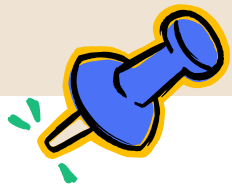


Width 與 Height 的 Max 與 Min

Width
Height } 固定

Max-
Min- } 可變動

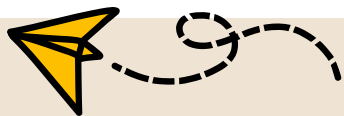


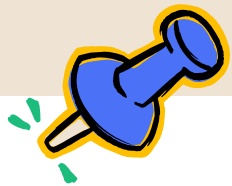


Width 與 Height 的 Max 與 Min

Max- { 最大
可小，不能再大

Min- { 最小
可大，不能再小





Width 與 Height 的 Max 與 Min

max-width

常用

min-width

min-height

overflow: auto ;

max-height

不常用

